
Firm Location Choice

LSE GY457 - AT 2025

Topic 6

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Course overview

- Block 1

- Intro to Urban and Regional Economics and course overview
- Topic 1: Regional and urban concentration forces
- Topic 2: The empirics of agglomeration
- Topic 3: The costs and benefits of agglomeration

- Block 2

- Topic 4&5: Monocentric city model
- **Topic 6: Firm location choice**

- Block 3

- Topic 7: The vertical dimension of cities
- Topic 8: Neighborhoods and segregation
- Topic 9: Gentrification and suburbanization
- Topic 10: Measuring cities with new technologies

Recap from last time

- Monocentric city model delivers useful predictions
 - Higher income leads to decentralization (when considering monetary commuting costs)
 - Better transportation infrastructure leads to decentralization
 - Higher agricultural land rent shrinks cities
- Some predictions validated with causal evidence
 - Cities decentralize with better transportation infrastructure
- Predictions useful for planners, developers, investors
 - Give insights into expected effects on demand, prices, densities

Today: Firm location choice

- Firms in the monocentric city model
 - Firm bid-rent
 - Land use segregation
- Agglomeration and decentralization
 - Endogenous agglomeration
 - Multiple equilibria
- Emergence of new clusters
 - Sub-centers
 - Edge cities
 - Historic anchoring

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Why do firms concentrate in the CBD?

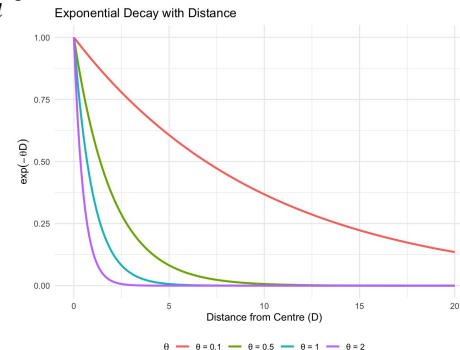
- To rationalize firm concentration in the CBD, we need a force that attracts them to the CBD
 - High cost of distance, so that firms outbid residents for most central locations
- Fundamental location factors (in the past)
 - Firms need to be close to natural harbor (trading cities)
 - Firms sell goods at a market place (von Thünen)
 - But transportation costs decreased over time: Plausible in the past
- Agglomeration forces declining with distance (today)
 - Being located centrally in a labor pool; knowledge spillovers
 - Knowledge spillovers very localized: More plausible today

Firm bid-rent: Framework

- Identical firms in services (e.g., bank, insurance) in location i maximize profits π_i using capital and land as inputs (Ahlfeldt and Wendland, 2013): $\pi_i = \chi_i K_i^\alpha \psi_i L_i$
 - We set price of good to 1 and price of capital normalized to 1
 - ψ_i is firm land rent
- Firms occupy land directly (no developers)
- Capital (broadly defined) includes workers, machines, computers, building structure, ...
 - Developers could be added (supplying office space to firms), analogously to what we did for the household bid-rent model

Firm bid-rent: Production function

- Assume Cobb-Douglas production function: $\chi_i = A_i K_i^\alpha L_i^{1-\alpha}$
 - Similar results with any concave production function ($\chi' > 0$, $\chi'' < 0$)
 - A_i productivity spillovers (location specific!)
 - $0 < \alpha < 1$ is capital share of inputs (constant in Cobb-Douglas)
- Let productivity spillovers A_i be a decreasing function of distance from CBD (D_i): $A_i = a_i e^{-\theta D_i}$
 - a_i is “fundamental production amenity” (does not change with distance from CBD, let's further assume is flat, i.e., $= a$)
 - $\theta > 0$ determines the decay of the spillover (see graph)



Firm bid-rent: Profit maximization problem

- Firms choose inputs to maximize profits:

$$\pi_i = A_i K_i^\alpha L_i^{1-\alpha} - K_i - \psi_i L_i$$

- First order conditions give:
 - Combining $\partial\pi_i/\partial K_i=0$, $\partial\pi_i/\partial L_i=0$: $K_i/L_i = [\alpha/(1-\alpha)]\psi_i$
- Capital density (K/L) increases in land rent
 - Similar to the developers maximization problem (Topic 4)

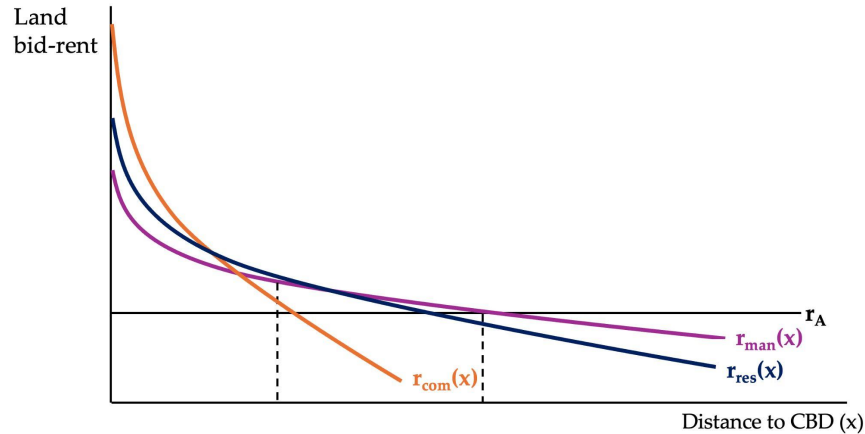
Firm bid-rent: Spatial equilibrium

- In spatial equilibrium, profits are equalized across locations
- Perfect competition implies zero profits
 - $\pi_i = A_i K_i^\alpha L_i^{1-\alpha} - K_i - \psi_i L_i = 0$
 - Symmetric condition to fixed utility in household bid-rent
- Given optimal choice of K, L and perfect competition:
 - $\pi_i(K_i^*, L_i^*) = 0$
 - Plug in first order condition: $\psi_i = [\alpha^\alpha (1-\alpha)^{1-\alpha}] A_i^{1/(1-\alpha)} = C e^{-\theta D_i / (1-\alpha)}$
 - Where we define constant $C = [\alpha^\alpha (1-\alpha)^{1-\alpha}] a^{1/(1-\alpha)}$
- $\partial \psi_i / \partial D_i < 0$
 - i.e., firm bid-rent decreases with distance from CBD
 - Higher office rents compensate for greater productivity near CBD

Today: Firm location choice

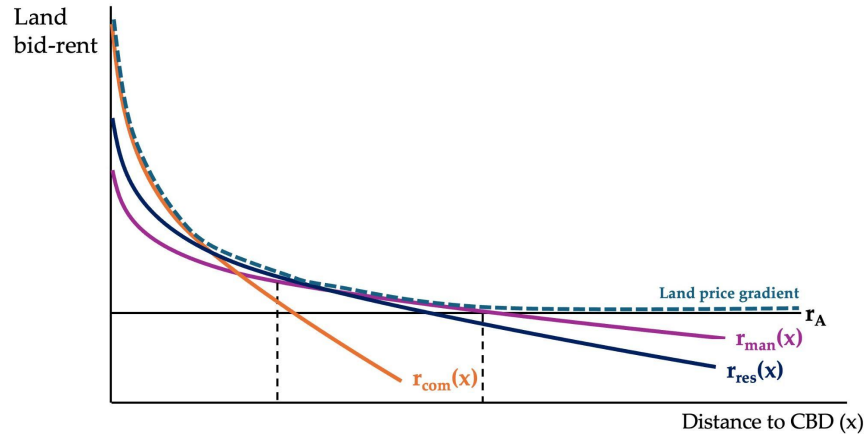
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Land price gradients by use



- Commercial bid-rent curve with large θ , r_{com}
- Residential bid-rent curve, r_{res}
- Manufacturing bid-rent curve with small θ , r_{man}
- Agricultural land rent, r_A
- Can we learn something about land use patterns?

Land price gradients by use

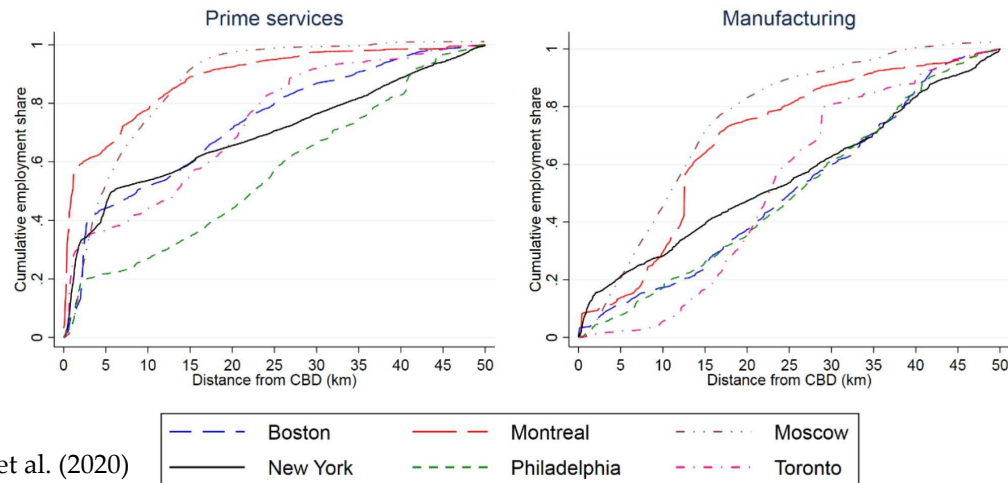


- Land goes to the highest bidder
- i.e. to the sector with the highest bid rent at a given distance from CBD
- Every bid-rent curve is convex
- Land price gradient even more convex due to *sorting* of different sectors in different locations

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CBDs dominated by prime services



Ahlfeldt et al. (2020)

- Prime services: Finance, insurance, real estate, rental, professional, ...
- Knowledge-based tradable services highly concentrated in CBD
- Manufacturing employment a lot more decentralized
- Location fundamentals not necessarily relevant for prime services

Adding bilateral productivity spillovers

- Assuming that an agglomeration economy emerges from the CBD is a (strong) simplification
- More realistic: Firms benefit from proximity to each other and hence cluster in the (endogenous) center
- To incorporate this idea in a model, the algebra soon gets quite more complex...
 - e.g. Ahlfeldt-Wendland (2013): $A_i = a_i e^{-\theta D_i} e^{\beta Z_i}$ where $Z_i = \sum_j \chi_j e^{-\tau d_{ij}}$
 - The extra term models productivity as depending on nearby firms output, weighted by distance (bidirectional agglomeration force)

CBD vs. bilateral spillovers in historical Berlin (Ahlfeldt and Wendland, 2013)

Tab. 1 **Industry Employment in 1890 and 1933 (Old-Berlin Boundaries)**

Area	Year	Manufacturing		Trade and Services	
		Employment	Employment Share	Employment	Employment Share
Total:	1890	313,799	42.22%	148,139	19.93%
	1933	329,352	35.80%	396,700	43.01%
CBD:	1890	117,556	27.11%	57,888	13.35%
	1933	91,931	34.30%	175,972	65.67%

- Transition from **historic** to **modern** CBD/economy
- Resembles many cities in developing world today
- Bilateral agglomeration should be increasingly important

CBD vs. bilateral spillovers in historical Berlin (Ahlfeldt and Wendland, 2013)

$$\log(\Psi_i) = \log(\alpha^\alpha) + \frac{\beta}{(1-\alpha)^2} \sum_j \Psi_j L_j e^{-\tau d_{ij}} - \frac{\theta}{1-\alpha} D_i - \sum_s \frac{\varphi_s}{1-\alpha} d_{is} \quad (9)$$

From the spatial equilibrium condition (9), some general implications regarding the city structure and its driving forces emerge. Land rents increase in access to economic activity and decrease in distance to the predefined historical center and locational amenities. The magnitudes of the marginal effects of all these location features increase in the share parameter of non-land inputs, which determines the rate of substitution between land and capital. This is particularly the case for the bidirectional decentralized agglomeration force, which is the intuitive result of surrounding firms substituting land for capital as the densities and agglomeration benefits increase.

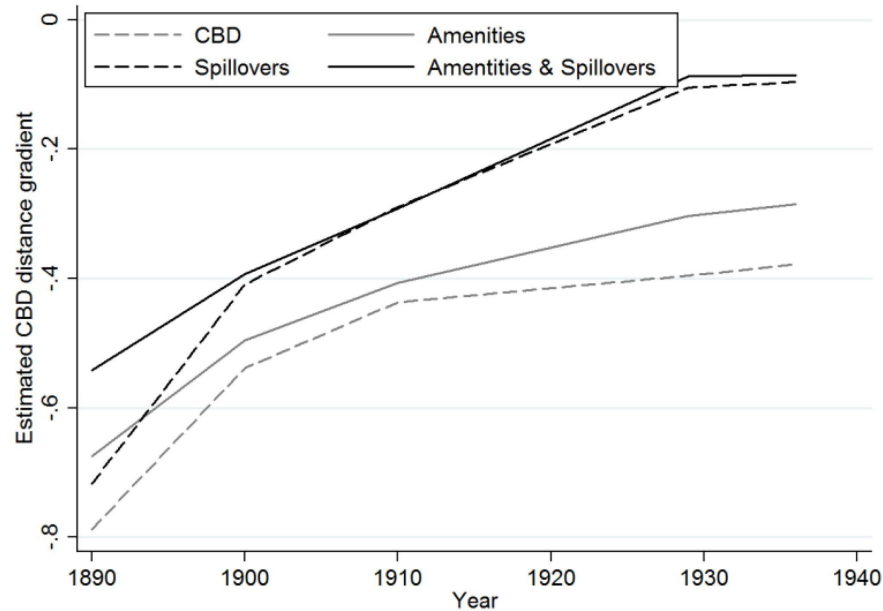
CBD vs. bilateral spillovers in historical Berlin (Ahlfeldt and Wendland, 2013)

- Empirical counterpart:

$$\log(NLV_{it}) = \varsigma_t + \varrho_t \left[\sum_j (S_j \times NLV_{jt}) e^{-\phi_t \times TT_{ijt}} \right] + \rho_t DCBD_i + \sum_s v_{ts} X_{sit} + \varepsilon_{it} \quad (12)$$

where NLV_i denotes the land value (adjusted to 2005 prices) at a given location, S is the surface area of the plot of land in commercial use (measured in square km), which is fixed over time, TT_{ij} are bilateral travel times measured in minutes and defined in equations (10 and 11) and X_s is a vector of hedonic location control variables. While ε_{it} is a random error term, all other Greek letters denote parameters to be estimated that find their correspondence in the spatial equilibrium equation (9). Equation (12) can be

CBD vs. bilateral spillovers in historical Berlin (Ahlfeldt and Wendland, 2013)



- What does the figure show?

CBD vs. bilateral spillovers in historical Berlin (Ahlfeldt and Wendland, 2013)

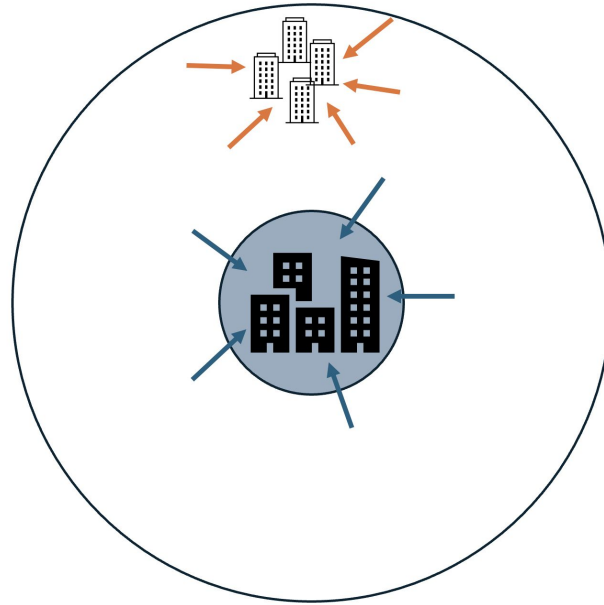
- Summary of results
- Between 1890-1940, (historical) city center very attractive for businesses
 - High concentration of agglomeration and amenities
- At the beginning of study period, bi-directional agglomeration economies (spillovers among firms) played relatively limited role vs. strong role of CBD
- Over time, importance of decentralized agglomeration forces grew vs. direct CBD effect diminished
 - As result of transitioning towards knowledge-based economy

Implications for urban structure: Strong CBD spillover



- New clusters may emerge in the periphery (e.g., start-ups) or shock relocates existing firms (e.g., after disaster)
- Exogenous CBD spillover pulls firms back into CBD
- City structure stable

Implications for urban structure: Bilateral agglomeration forces



- New clusters may emerge in the periphery (e.g., start-ups) or shock relocates existing firms (e.g., after disaster)
- Cluster generates own agglomeration economy
- City structure changes
- **Multiple equilibria!**

Causes of employment decentralization

- Manufacturing should decentralize given change in transportation costs, production, inventory technologies
- Tradable services could decentralize, given bilateral agglomeration forces allow for multiple equilibria
- Driving forces for employment decentralization:
 - Changes in transportation technology
 - Suburbanization of people (jobs follow people)
 - Fiscal and social problems
 - Ageing building stock
 - Regulation (height constraint, land use); Other?

Evidence: Glaeser and Kahn (2001)

- US cities mostly decentralized (NY is exception)
 - Share of employment within 3 miles of CBD is less than 29%
 - Employment share of central MSA areas falling since 1950s
- Hard to predict decentralization, but:
 - Manufacturing particularly decentralized
 - Suburbanization associated with decentralized employment
 - Cities specialized in services are relatively more centralized
 - Knowledge-based industries more likely to locate in center
- Evidence supports theoretical expectations

Evidence: Baum-Snow et al. (2017)

- Transportation infrastructure improvements between 1990-2010 contributed to decentralization of population and employment in China
 - Radial highways decentralize services activity
 - Radial railroads decentralize 20% of industrial activity
 - Ring roads decentralize both (50% of industrial activity)
 - Radial/ring roads displace 4%/20% of central city population
- Identification strategy similar to Baum-Snow (2007)
 - Seen in Topic 5

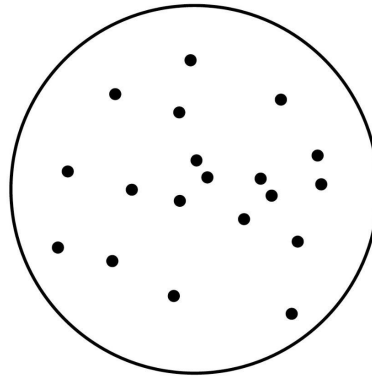
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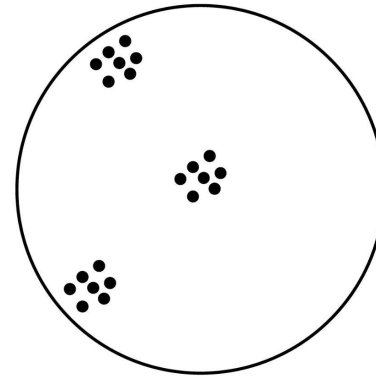
Decentralized vs. dispersed employment

- Employment decentralized in many cities
- Not necessarily the same as dispersed

Decentralized and dispersed



Decentralized and clustered



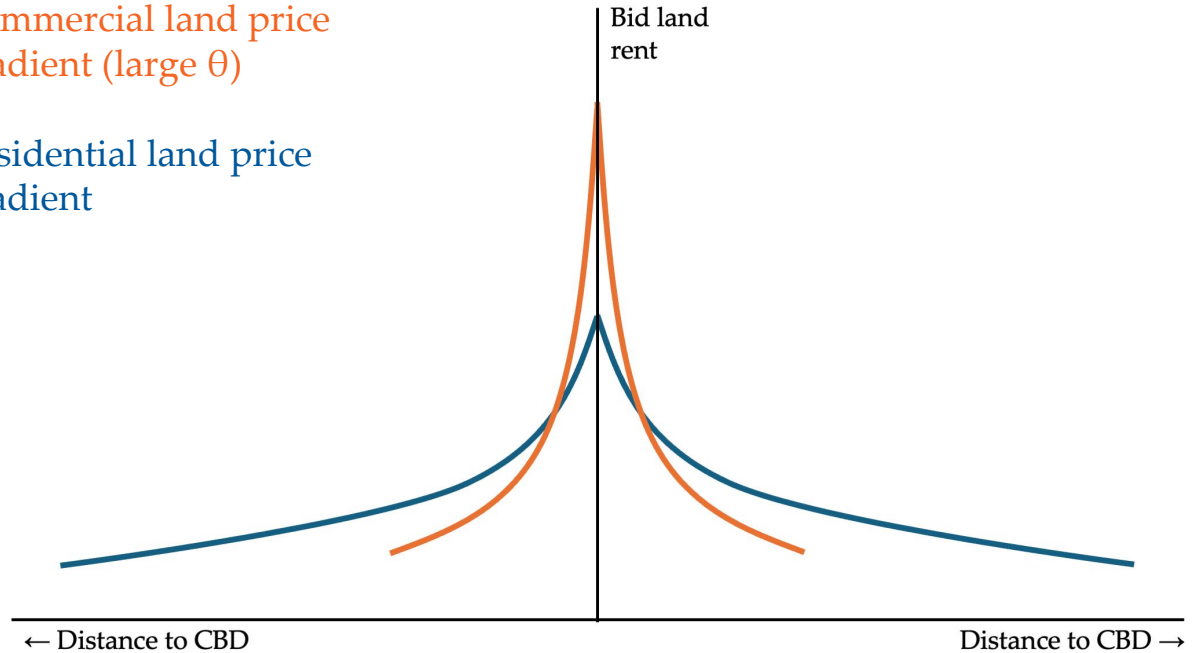
Why, when, and where do cities develop firm clusters outside the CBD?

- Monocentric vs. polycentric cities
- (Dis)advantage of large monocentric cities
 - Large agglomeration economies in CBD
 - Long commuting costs and higher wages as compensation
 - High land prices
- Large polycentric cities
 - Have one or multiple subcenters
 - Subcenters are employment clusters similar to CBD
 - Offer some agglomeration benefits
 - Reduce commuting costs and land prices
 - Polycentric cities may combine benefits of large and small cities

Large monocentric city

Commercial land price
gradient (large θ)

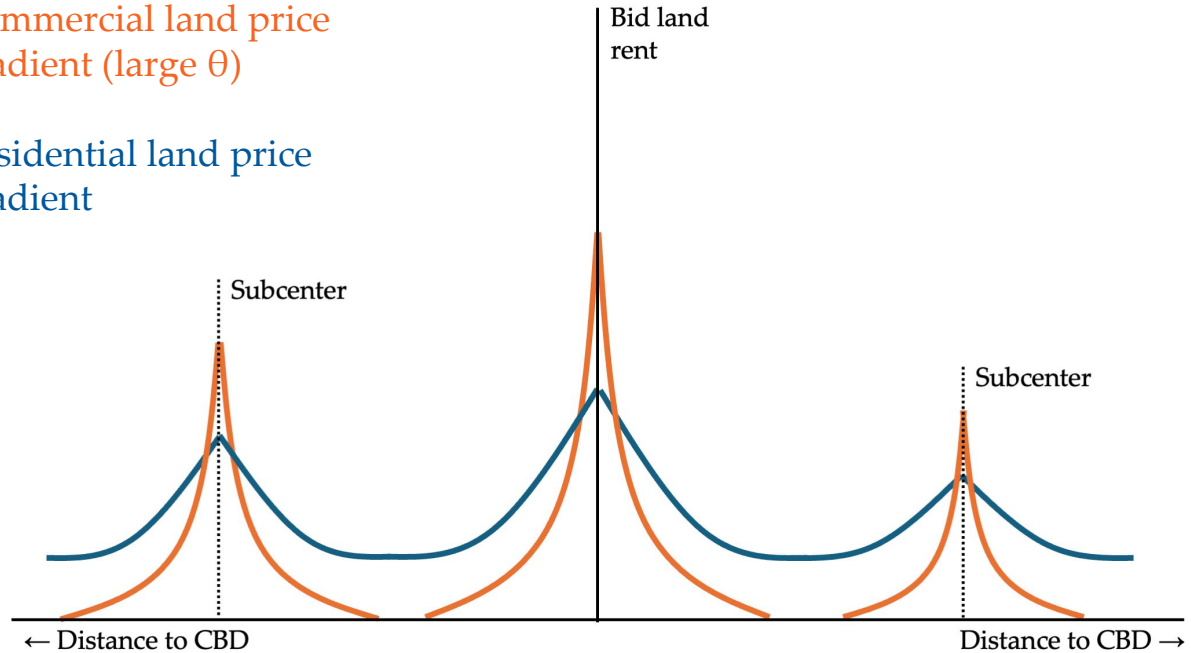
Residential land price
gradient



Large polycentric city

Commercial land price
gradient (large θ)

Residential land price
gradient



When do cities develop sub-centers?

- Fujita and Ogawa (1982) model
 - Number of subcenters increases with (i) population and (ii) commuting costs
 - Sub-centers emerge endogenously to keep the city efficient
- McMillen and Smith (2003) test the predictions for US
 - Identify sub-centers as peaks in local employment densities
 - Run Poisson regressions (count models) to analyze the determinants of the number of sub-centers per city
 - How do city size, congestion, and other factors influence the number of sub-centers in US cities?

Identifying sub-centers in the data

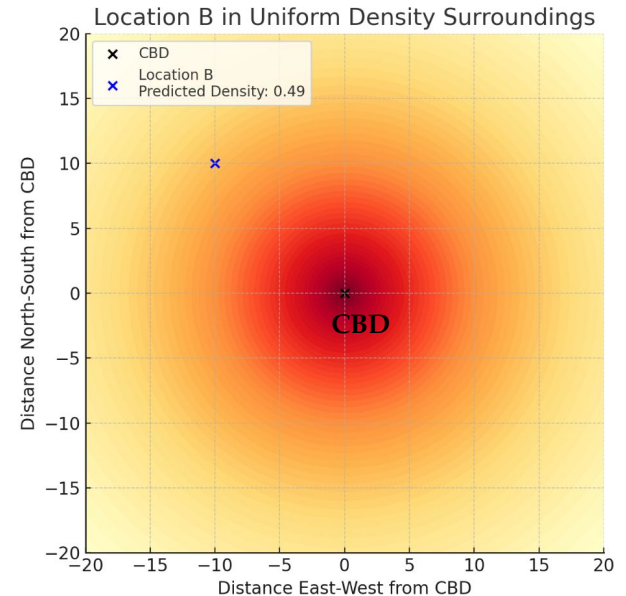
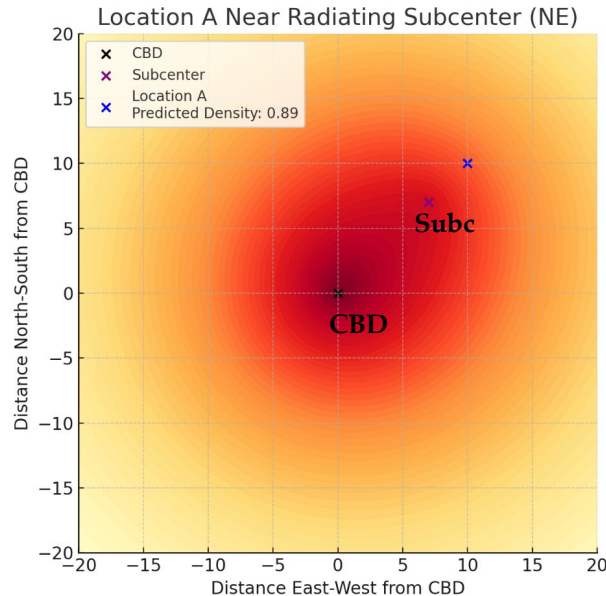
(McMillen and Smith, 2003)

- Idea: Identify local peaks in employment density within a metropolitan area
- First step: Regress (log) employment density on distance from the CBD
 - Model creates an “expected” employment density surface across the city, based on proximity to the CBD
 - Allows to inspect unusually high-density locations that may represent sub-centers
 - In calculating this predicted values of employment density, more weight is given to nearby locations (creating smooth density surface)

Identifying sub-centers in the data

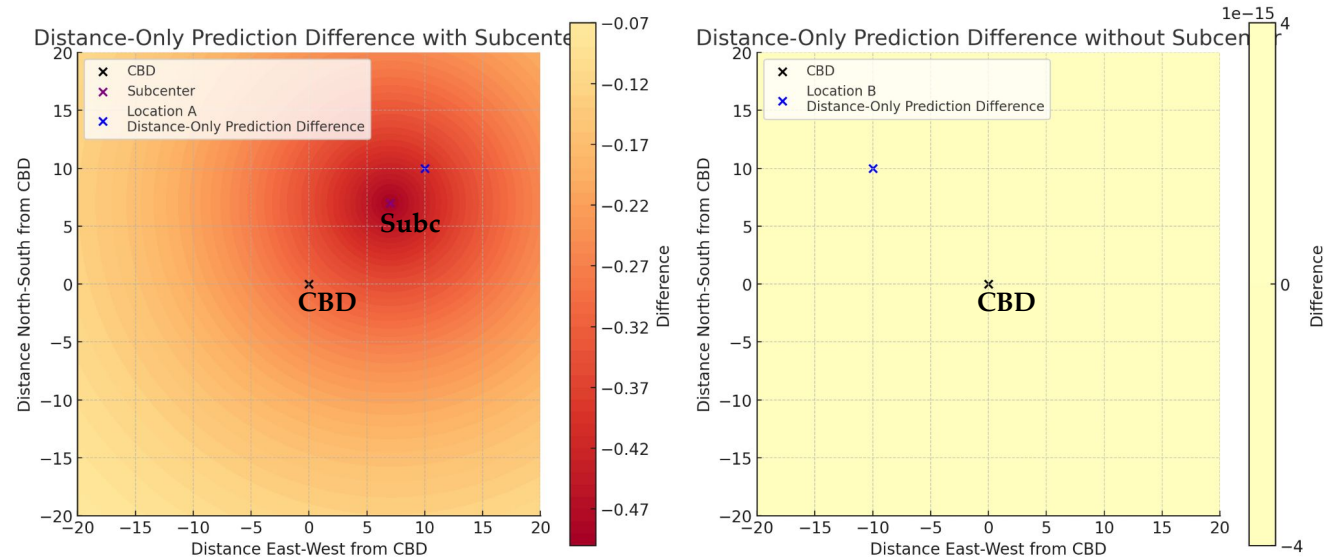
(McMillen and Smith, 2003)

Plot: Location A and B at same distance from CBD, but A close to sub-center



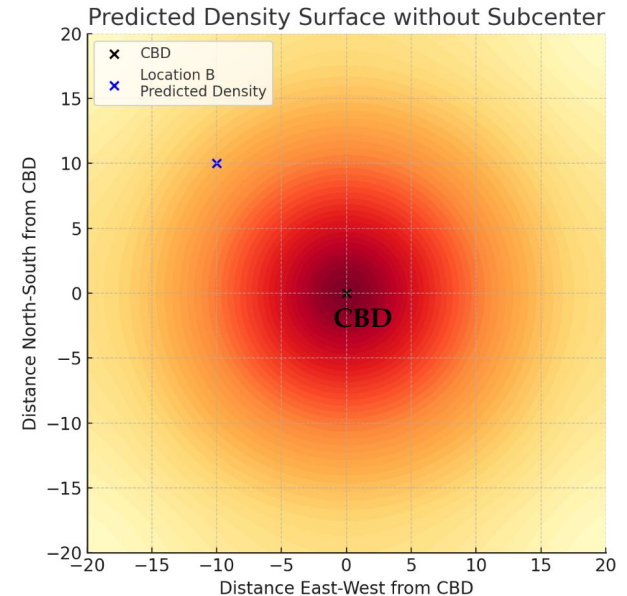
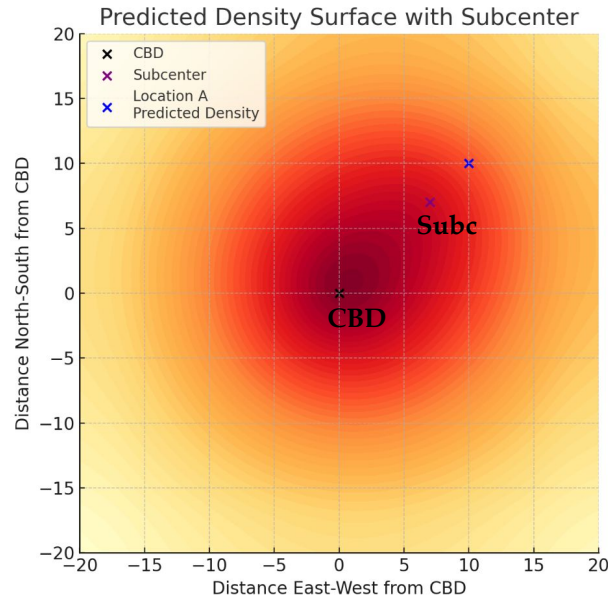
Identifying sub-centers in the data (McMillen and Smith, 2003)

Plot: Distance-only employment density – Actual density



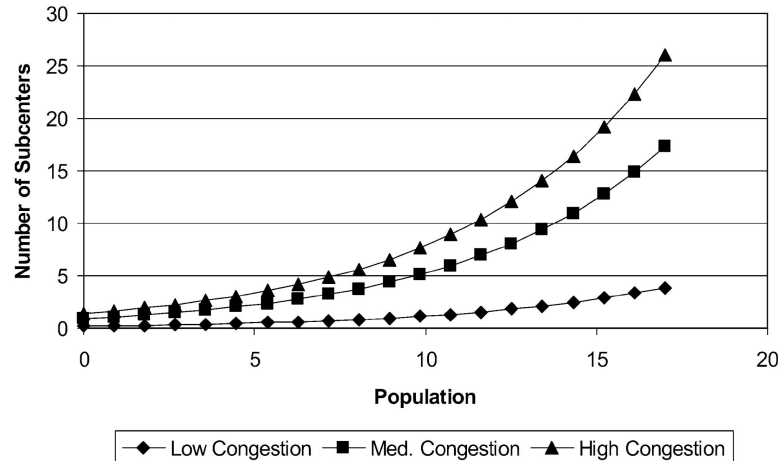
Identifying sub-centers in the data (McMillen and Smith, 2003)

Plot: Predicted density by distance to CBD and weighting by nearby density



Determinants of sub-center formation

(McMillen and Smith, 2003)



- At low population levels, additional congestion has little impact on expected number of sub-centers
- Higher levels of population and/or congestion rapidly increase expected number of sub-centers

Determinants of sub-center formation

(McMillen and Smith, 2003)

- Additional results:
 - Cities with older housing stock tend to have more subcenters
 - Age of the central city (defined as number of years since the central city first reached 25% of its 1990 population), median income, and proportion of metropolitan population in central city are statistically not different from zero across all specifications
 - Did you expect different effects? Other variables not included that may help explaining sub-center formation?

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Edge cities

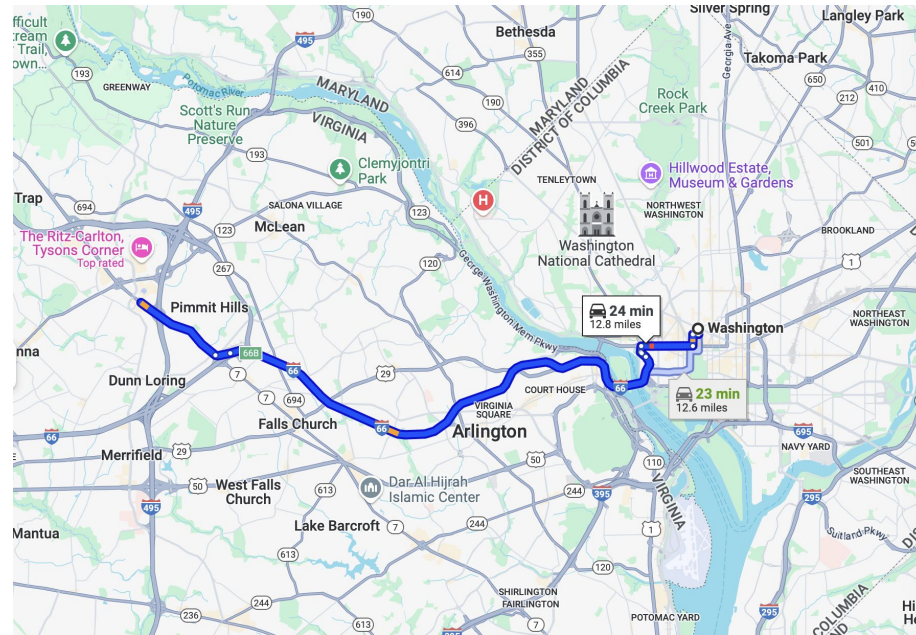
- Where would a profit maximizing developer strategically develop a satellite city?
- Term popularized by WP journalist/author Joel Garreau
 - 1991 book titled “Edge city: Life on the new frontier”
- New cities in urban periphery built by one developer
 - Built to host knowledge based tradable services
 - Dominated by class A office stock (shiny offices)
 - Net commuting destination (unlike residential suburb)
 - Has more jobs than bedrooms
 - Proper city with employment, recreation, and entertainment
- Developed by a single developer (vs. sub-center)
 - Examples: Canary Wharf in London, La Défense district in Paris

Why create edge cities?

- Developer goals to create an attractive edge city:
 - Internalize agglomeration spillovers (e.g., maximize productivity)
 - Avoid social and fiscal problems (poverty, crime, ...)
 - Reducing capacity constraints (like congestion)
- Typical tools and strategies:
 - Mixed-use spaces to minimize commuting distances
 - Optimal road layout and parking capacity (adapted to demand)
 - Non-distortionary taxes (limited redistribution to keep incentives)
- Developer solves a “coordination problem” inherent to agglomeration
 - Value of land low pre-development, but high afterwards: Profits!

Archetypal edge city: Tyson's Corner

(developed by Til Hazel, of Hazel & Peterson)



Archetypal edge city: Tyson's Corner



- Largest retail area on the East Coast south of NY
- 3400 hotel rooms, 100,000+ jobs, over 28 million sq ft A-level office space
 - Whereas 17 million in Washington city

Henderson and Mitra (1996)

- Edge city developed by a profit maximizing developer
 - Returns from producing tradable services good
 - Faces costs of providing space (fixed cost of capital p_k)
 - Faces cost of hiring labor: Wages increase in (i) residential rent (compensation); (ii) number of workers in metro area (congestion)
- Developer chooses
 - Capacity K of edge city (floor space)
 - Workforce in the edge city
 - Location of the edge city in terms of distance from port city (CBD)
- Developer has monopsony power

Henderson and Mitra (1996)

- Port city (CBD) and edge city generate agglomeration economies which depend on
 - Population of port city and of the edge city
 - Spillovers to each other, which decrease with distance
- Developer chooses location of edge city, facing tradeoff
- Locating the edge city closer to the port city (CBD):
 - Increases cost of land, need higher compensation for workers
 - Productivity increases, due to larger spillovers from port city
 - Port city productivity and competitiveness increase

Henderson and Mitra (1996)

- Result: Small changes in CBD capacity K_0 can trigger large effects on the location of edge city
 - Higher CBD capacity K_0 reduces developer monopsony power (CBD pays higher wages given larger agglomeration effects)
 - Developers need to consider K_0 when choosing edge city capacity
 - In tendency, larger port cities imply smaller edge cities
- Model highly sensitive to parameter values (“everything goes”)
 - *Chaotic* patterns resemble Garreau’s notion of *randomness*
 - But: chaotic and randomness not the same

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Historic anchoring

- Why some cities did not decentralize?
- e.g. comparing NYC to LA:
 - LA has many more remote prime locations relative to NYC, despite similar population size today
- Role of history?
 - NYC population in 1900: 3.4 million; LA population in 1900: 100k
 - NY built a much larger transit history, much earlier
 - “Whereas the old downtown garment district had been anchored by the value of proximity to the port, Lefcourt’s new garment district lay between Pennsylvania and Grand Central stations, anchored by the rail lines, which continued to give New York a transportation advantage. Transportation technologies shape cities, and Midtown Manhattan was built around two great rail stations that could carry in oceans of people.” (Glaeser, 2011)

Historic anchoring

- Larger cities in the 1900 more likely to develop extensive transit system (Ahlfeldt et al., 2020)
 - Naturally oriented towards the city core
 - Reinforces a city's locational advantage and makes subsequent changes in the location of prime services less likely
 - Economic activity more likely to remain “anchored” close to the historical CBD
- Can help explaining differences in city structure between US east and west coast, US vs. European/Asian cities

Summary

- Firm bid-rent can be derived using zero-profit condition
 - Historically, high transportation costs relevant
 - Agglomeration spillovers more plausible determinant today
- Changes in transport technology led to employment decentralization
 - Manufacturing employment decentralizes due to lower transportation costs
 - Services can decentralize since agglomeration is endogenous
- Decentralized vs. dispersed employment
 - Subcenters can emerge endogenously if cities get larger
 - Developers strategically build edge cities to make profits

Next week: Vertical dimension of cities

- Model
 - Vertical and horizontal costs and benefits
 - General equilibrium
- Do rents increase in height within buildings?
 - Returns to height
 - Cost of height
- Equilibrium and counterfactuals
 - Urbanization as a cause of skyscraper development
 - Skyscraper development as cause of urbanization
 - Height regulation

Thank you!

Readings and references

Required readings

- Ahlfeldt, G. M., and Wendland, N. (2013). How polycentric is a monocentric city? Centers, spillovers and hysteresis. *Journal of Economic Geography*, 13(1), 53-83.
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Other readings and references

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